

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \quad z_i \in \mathbb{C}^n$$

$z^T \cdot z$ is no good for length
Example,

$$z = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$z^T \cdot z = 0$$

So, we use

$$\bar{z}^T \cdot z = z^H \cdot z$$

$$\begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$Q \bar{Q}^T = I$$

$$Q Q^H = I$$

Fourier Matrix

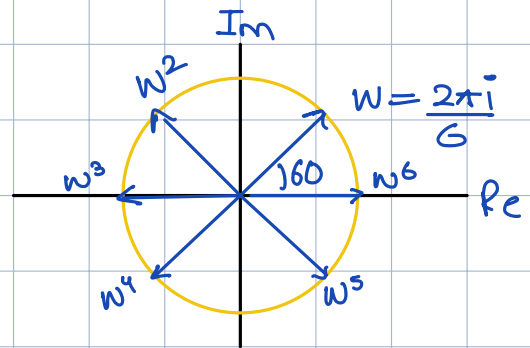
$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & & \omega^{2(n-1)} \\ \vdots & \vdots & & & \vdots \end{bmatrix}$$

$$\begin{bmatrix} 1 & W^{(n-1)} & W^{2(n-1)} & W^{(n-1)^2} \end{bmatrix}_{n \times n}$$

$$(F_n)_{ij} = W^{ij} \quad i, j = 0 \dots n-1$$

$$W^n = 1 \quad W = e^{i 2\pi/n}$$

$$= \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$



Now,

$$W = e^{2\pi i / 4} = 90^\circ = i \quad i, i^2 = -1, i^3 = -i, i^4 = 1$$

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & -i & -1 & i \end{bmatrix}$$

Columns are orthogonal

$$F_4^H F = I$$

$$(W_{64})^2 = W_{32}$$

$$\begin{bmatrix} F_{64} \end{bmatrix} = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_{32} & 0 \\ 0 & F_{32} \end{bmatrix}$$

↳ Permutation matrix

Even - odd

$$64^2 = 2(32) + 32$$

$$D = \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & n \end{bmatrix}$$

permutation
 Picks first even
 $x_0, x_2, \dots$
 then odd
 $x_1, x_3, \dots$